In [1]:
```
import numpy as np
```

In [2]:
```
np.__version__
```

Out[2]:
```
'2.3.5'
```

In [3]:
```
my_list = [0,1,2,3,4,5]
my_list
```

Out[3]:
```
[0, 1, 2, 3, 4, 5]
```

In [4]:
```
type(my_list)
```

Out[4]:
```
list
```

In [5]:
```
arr = np.array(my_list)   # converting list to array
arr
```

Out[5]:
```
array([0, 1, 2, 3, 4, 5])
```

In [6]:
```
type(arr)
```

Out[6]:
```
numpy.ndarray
```

In [8]:
```
np.arange(10)  #arange it will print the value within interval
```

Out[8]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [11]:
```
np.arange(10,15)
```

Out[11]:
```
array([10, 11, 12, 13, 14])
```

In [9]:
```
np.arange(10,20,5)
```

Out[9]:
```
array([10, 15])
```

In [13]:
```
np.arange(10,20,5,2)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[13], line 1
----> 1 np.arange(10,20,5,2)

TypeError: Cannot interpret '2' as a data type
```

In [10]:
```
np.zeros(5) #parameter tunning
```

Out[10]:
```
array([0., 0., 0., 0., 0.])
```

In [15]:
```
np.zeros(5, dtype=int) #hyperparameter tunning
```

Out[15]:
```
array([0, 0, 0, 0, 0])
```

In [16]:
```
np.zeros([2,2], dtype = int) #2d array
```

Out[16]:
```
array([[0, 0],
       [0, 0]])
```

In [17]:
```
np.zeros([5,10], dtype = int)
```

Out[17]:
```
array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

In [11]:
```
np.ones(3)
```

Out[11]:
```
array([1., 1., 1.])
```

In [13]:
```
np.ones([3,5], dtype = int)
```

Out[13]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [14]:
```
np.ones([3,5])
```

Out[14]:
```
array([[1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.]])
```

In [15]:
```
np.arange(10,20)
```

Out[15]:
```
array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

In [16]:
```
np.arange(20, 10)
```

Out[16]:
```
array([], dtype=int64)
```

In [17]:
```
np.arange(-20, 10)
```

Out[17]:
```
array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10,  -9,  -8,
        -7,  -6,  -5,  -4,  -3,  -2,  -1,   0,   1,   2,   3,   4,   5,
         6,   7,   8,   9])
```

In [18]:
```
np.twos(3)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[18], line 1
----> 1 np.twos(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

In [29]:
```
np.threes(3)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[29], line 1
----> 1 np.threes(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'threes'
```

In [19]:
```
random.rand(3,2)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[19], line 1
----> 1 random.rand(3,2)

NameError: name 'random' is not defined
```

In [21]:
```
np.random.rand(2)
```

Out[21]:
```
array([0.12612996, 0.31317959])
```

In [22]:
```
np.random.rand(5)
```

Out[22]:
```
array([0.34640935, 0.71572261, 0.37985668, 0.08954054, 0.03836129])
```

In [23]:
```
np.random.rand(3,2)
```

Out[23]:
```
array([[0.55524513, 0.88119264],
       [0.84528654, 0.53136557],
       [0.88056207, 0.32686293]])
```

In [26]:
```
np.random.randint(3)
```

Out[26]:
```
0
```

In [29]:
```
np.random.randint(5,10)
```

Out[29]:
```
8
```

In [72]:
```
np.random.randint(5,10,3)
```

Out[72]:
```
array([9, 7, 7], dtype=int32)
```

In [73]:
```
ran = np.random.randint(5,10,3)
print(ran)
```

```
[8 9 6]
```

In [30]:
```
otp = np.random.randint(5,10,4)
print(otp)
```

```
[8 8 6 9]
```

In [31]:
```
arr0 = np.random.randint(10,40,(10,10))
print(arr0)
```

```
[[17 28 16 35 37 10 14 33 39 15]
 [13 34 14 24 36 32 37 17 35 13]
 [27 11 22 10 12 25 29 37 27 29]
 [16 13 34 20 35 28 22 20 37 29]
 [31 22 10 32 11 19 35 17 26 16]
 [32 14 13 36 12 27 18 20 32 39]
 [16 16 29 11 27 19 26 27 37 10]
 [13 30 36 17 16 38 19 36 25 34]
 [39 33 11 30 35 11 24 16 20 10]
 [34 27 37 11 13 23 28 23 24 23]]
```

In [32]:
```
arr0
```

Out[32]:
```
array([[17, 28, 16, 35, 37, 10, 14, 33, 39, 15],
       [13, 34, 14, 24, 36, 32, 37, 17, 35, 13],
       [27, 11, 22, 10, 12, 25, 29, 37, 27, 29],
       [16, 13, 34, 20, 35, 28, 22, 20, 37, 29],
       [31, 22, 10, 32, 11, 19, 35, 17, 26, 16],
       [32, 14, 13, 36, 12, 27, 18, 20, 32, 39],
       [16, 16, 29, 11, 27, 19, 26, 27, 37, 10],
       [13, 30, 36, 17, 16, 38, 19, 36, 25, 34],
       [39, 33, 11, 30, 35, 11, 24, 16, 20, 10],
       [34, 27, 37, 11, 13, 23, 28, 23, 24, 23]], dtype=int32)
```

In [39]:
```
arr1 = np.random.randint(1,5,(4,3))
print(arr1)
```

```
[[2 3 2]
 [2 4 3]
 [4 3 4]
 [1 3 1]]
```

, indexing, slicing in array

In [40]:
```
arr1[:]
```

Out[40]:
```
array([[2, 3, 2],
       [2, 4, 3],
       [4, 3, 4],
       [1, 3, 1]], dtype=int32)
```

In [42]:
```
len(arr1) #
```

Out[42]:
```
4
```

In [44]:
```
arr1.shape
```

Out[44]:
```
(4, 3)
```

In [45]:
```
arr0.shape
```

Out[45]:
```
(10, 10)
```

In [46]:
```
len(arr0)
```

Out[46]:
```
10
```

In [47]:
```
arr1
```

Out[47]:
```
array([[2, 3, 2],
       [2, 4, 3],
       [4, 3, 4],
       [1, 3, 1]], dtype=int32)
```

In [50]:
```
arr1[-2] # print row
```

Out[50]:
```
array([4, 3, 4], dtype=int32)
```

In [51]:
```
arr1[1]  # print row
```

Out[51]:
```
array([2, 4, 3], dtype=int32)
```

In [52]:
```
arr1
```

Out[52]:
```
array([[2, 3, 2],
       [2, 4, 3],
       [4, 3, 4],
       [1, 3, 1]], dtype=int32)
```

In [55]:
```
arr1[:,-1]
```

Out[55]:
```
array([2, 3, 4, 1], dtype=int32)
```

In [56]:
```
arr1[:,2]
```

Out[56]:
```
array([2, 3, 4, 1], dtype=int32)
```

In [57]:
```
arr1
```

Out[57]:
```
array([[2, 3, 2],
       [2, 4, 3],
       [4, 3, 4],
       [1, 3, 1]], dtype=int32)
```

In [58]:
```
arr1[0]
```

Out[58]:
```
array([2, 3, 2], dtype=int32)
```

In [59]:
```
v = arr1[1, 2]
print(v)
```

```
3
```

In [62]:
```
arr0
```

Out[62]:
```
array([[17, 28, 16, 35, 37, 10, 14, 33, 39, 15],
       [13, 34, 14, 24, 36, 32, 37, 17, 35, 13],
       [27, 11, 22, 10, 12, 25, 29, 37, 27, 29],
       [16, 13, 34, 20, 35, 28, 22, 20, 37, 29],
       [31, 22, 10, 32, 11, 19, 35, 17, 26, 16],
       [32, 14, 13, 36, 12, 27, 18, 20, 32, 39],
       [16, 16, 29, 11, 27, 19, 26, 27, 37, 10],
       [13, 30, 36, 17, 16, 38, 19, 36, 25, 34],
       [39, 33, 11, 30, 35, 11, 24, 16, 20, 10],
       [34, 27, 37, 11, 13, 23, 28, 23, 24, 23]], dtype=int32)
```

In [63]:
```
arr0[-7]
```

Out[63]:
```
array([16, 13, 34, 20, 35, 28, 22, 20, 37, 29], dtype=int32)
```

In [64]:
```
arr0[:, -7]
```

Out[64]:
```
array([35, 24, 10, 20, 32, 36, 11, 17, 30, 11], dtype=int32)
```

In [65]:
```
arr0
```

Out[65]:
```
array([[17, 28, 16, 35, 37, 10, 14, 33, 39, 15],
       [13, 34, 14, 24, 36, 32, 37, 17, 35, 13],
       [27, 11, 22, 10, 12, 25, 29, 37, 27, 29],
       [16, 13, 34, 20, 35, 28, 22, 20, 37, 29],
       [31, 22, 10, 32, 11, 19, 35, 17, 26, 16],
       [32, 14, 13, 36, 12, 27, 18, 20, 32, 39],
       [16, 16, 29, 11, 27, 19, 26, 27, 37, 10],
       [13, 30, 36, 17, 16, 38, 19, 36, 25, 34],
       [39, 33, 11, 30, 35, 11, 24, 16, 20, 10],
       [34, 27, 37, 11, 13, 23, 28, 23, 24, 23]], dtype=int32)
```

In [66]:
```
arr0[3:4, 2:6]
```

Out[66]:
```
array([[34, 20, 35, 28]], dtype=int32)
```

In [67]:
```
arr1
```

Out[67]:
```
array([[2, 3, 2],
       [2, 4, 3],
       [4, 3, 4],
       [1, 3, 1]], dtype=int32)
```

In [68]:
```
arr1[1]
```

Out[68]:
```
array([2, 4, 3], dtype=int32)
```

In [69]:
```
arr1[1,2] # 1 first row & 2 means colums
```

Out[69]:
```
np.int32(3)
```

In [70]:
```
arr1
```

Out[70]:
```
array([[2, 3, 2],
       [2, 4, 3],
       [4, 3, 4],
       [1, 3, 1]], dtype=int32)
```

In [71]:
```
arr1[1:2]
```

Out[71]:
```
array([[2, 4, 3]], dtype=int32)
```

In [ ]: